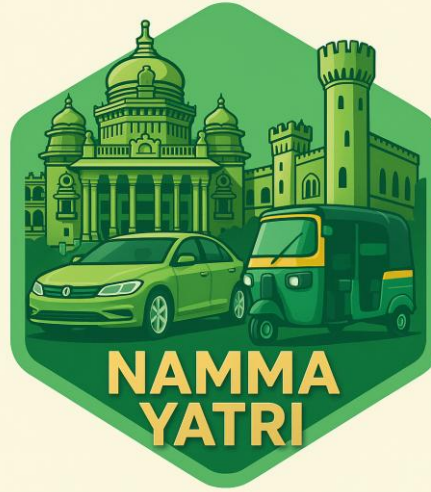




Optimizing Marketing Strategies – Improving Operational Efficiency and Revenue generation

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Group Name: ALRR0625



Agenda

- 1 Objective & Goals
- 2 Findings
- 3 Background
- 4 Insights and Recommendations
- 5 Recommendations
- 6 Appendix



Overview & Objectives

Business Overview

Namma Yatri plays a pivotal role in Bengaluru's urban transport network. Operating in a dynamic environment, the company must continuously adapt to shifting demand patterns, optimize ride allocation, and maintain profitability — all while delivering a superior customer experience. To support this, Namma Yatri aims to deepen its understanding of ride patterns, customer behavior, and operational bottlenecks.

Objectives

The objective is to uncover insights that can guide decisions to improve service efficiency, maximize revenue, and enhance customer satisfaction through:

- ❑ Identifying peak demand patterns and trends
- ❑ Uncovering factors influencing cancellations and driver performance
- ❑ Optimizing payment methods and trip durations
- ❑ Improving resource allocation and marketing strategies



Background – Initial observations

Background

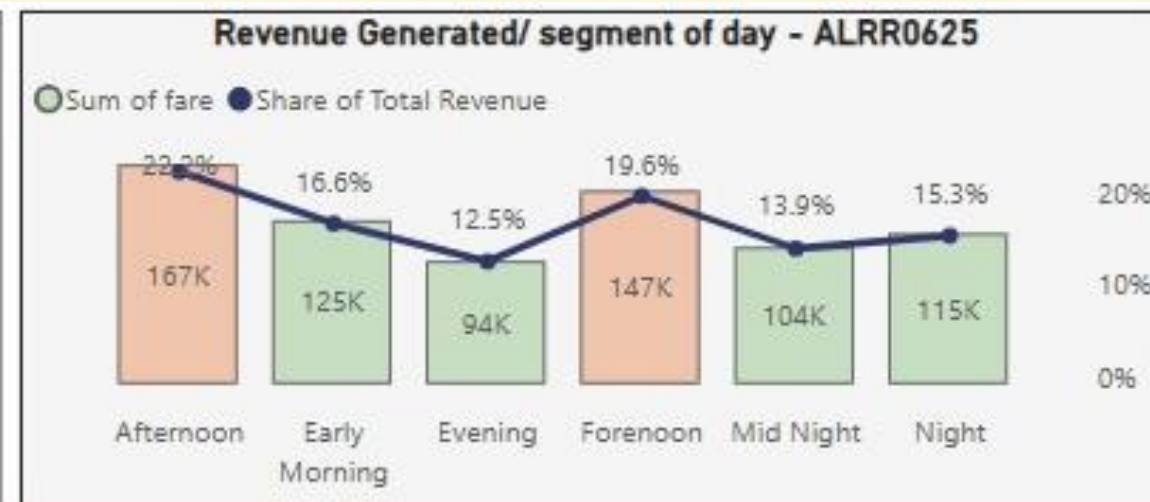
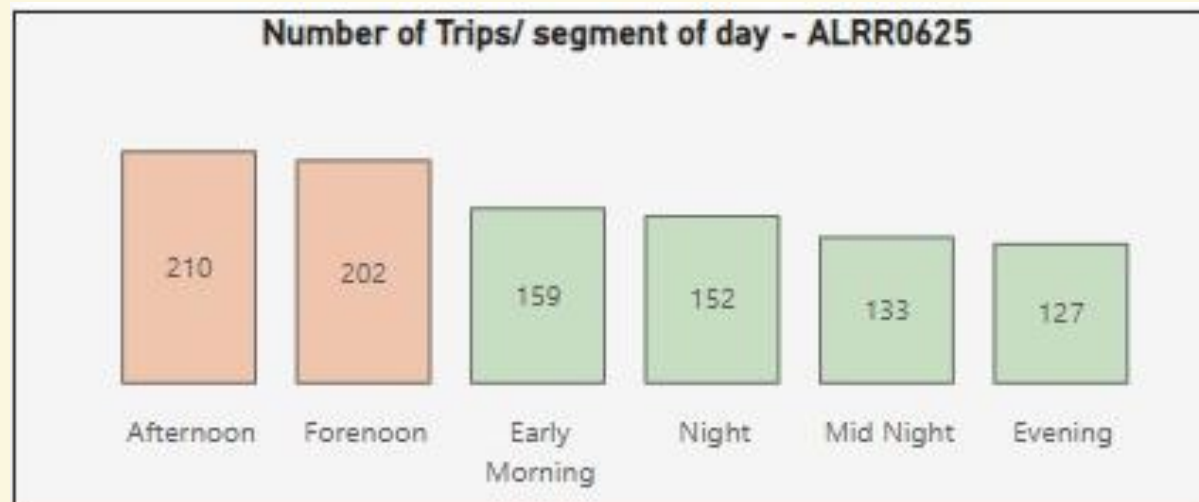
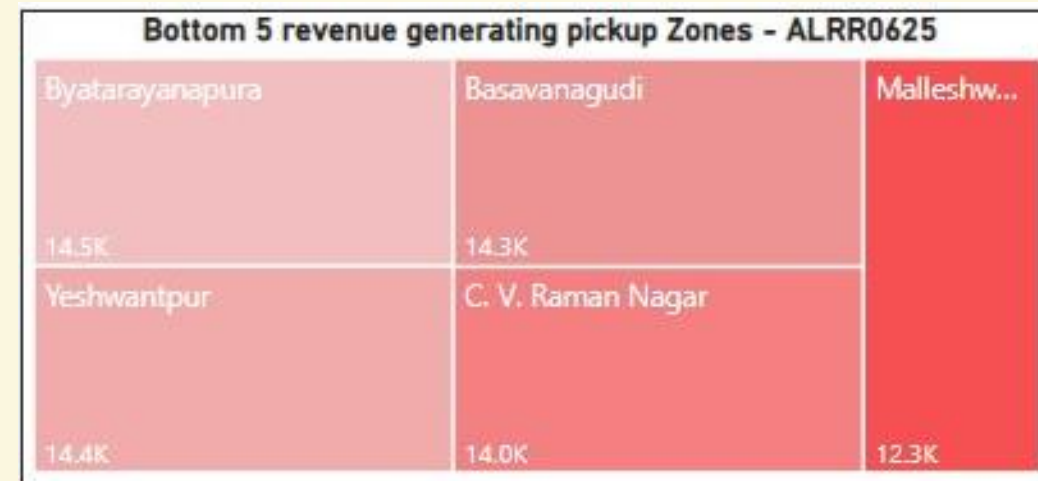
- ❑ Peak demand patterns and trends – Afternoon and Forenoon
- ❑ Cancellation rate stands at 23%
- ❑ Driver performance – 88%
- ❑ Payment method – Digital payments 74%
- ❑ Low Conversion rate – 45%



42% of revenue is generated during business hours

Insights

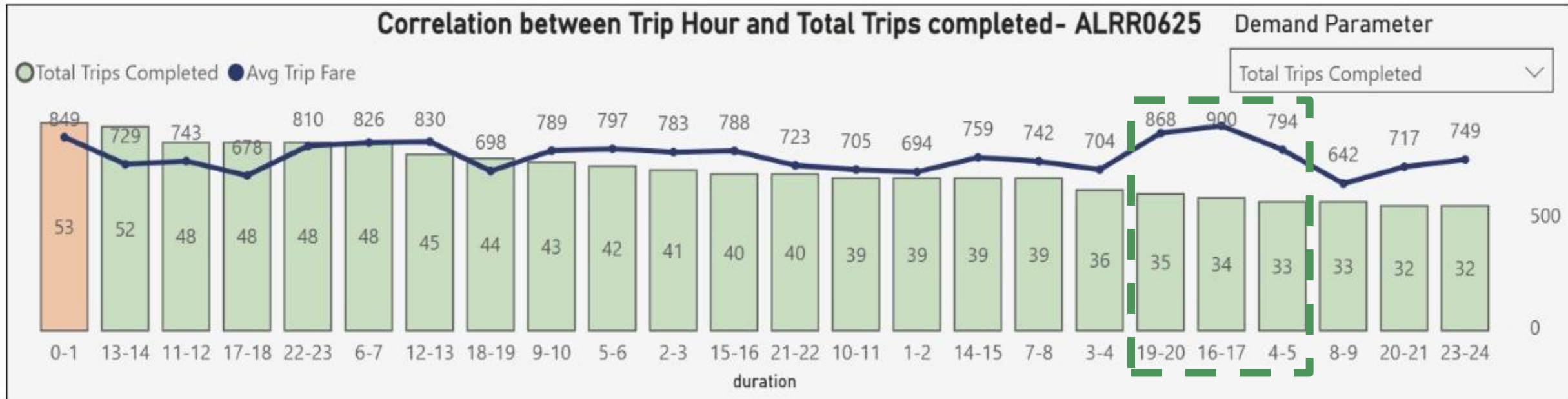
- Maximum trips & revenue are between **7 AM to 5 PM**
- Time slot Midnight to **1 AM** has the highest hourly revenue – **45K**
- Bottom 5 pickup zones contribute only **9.3% (69.5K)** of total revenue



High fare duration slots remain underutilized

Insights

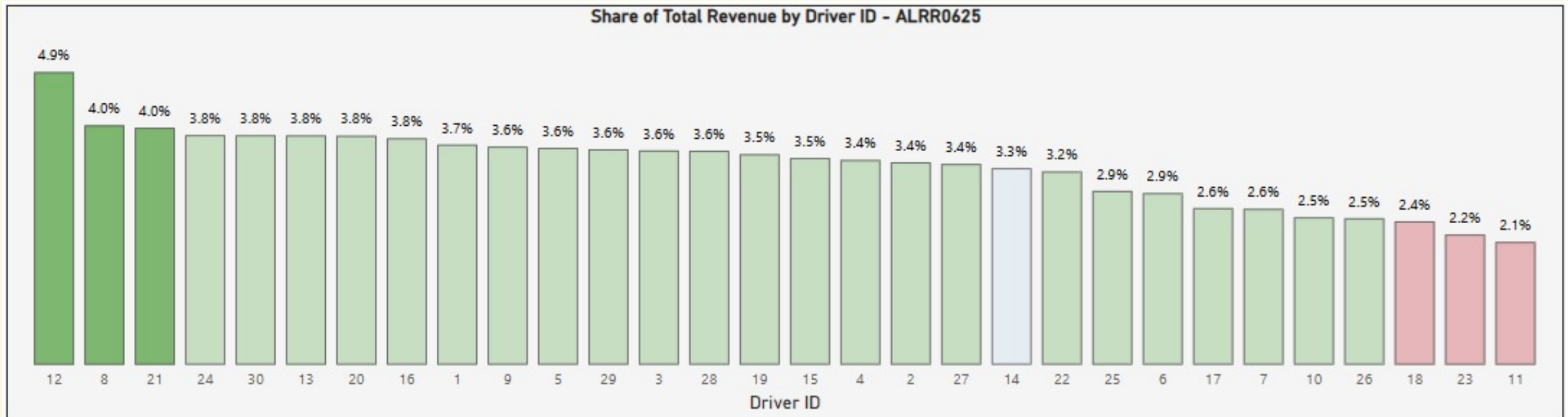
- ❑ Top 5 durations - average fare wise: - 16-17, 19-20, 0-1, 6-7, 22-23
- ❑ Early Morning Peak: 0-1 hour (12 AM - 1 AM) has the highest trip completion (849).
- ❑ Stable Fares: Average trip fare remains consistently similar regardless of trip volume.
- ❑ Mid-Day/Late-Night Dips: Trip completion is lowest during mid-day and late-night hours.
- ❑ Fare Spikes During Off-Peak Hours: While overall fares are stable, the "Avg Trip Fare" experiences noticeable increases during 16-17, 19-20, and 4-5 hours. These are times when "Total Trips Completed" are relatively lower compared to the peak, suggesting potential surge pricing or higher value trips during these less busy periods.



33% of our drivers are performing below average

Insights

- ❑ Average Revenue per driver – **3.3% (25K)**
- ❑ Drivers below average revenue – **33% (10)**
- ❑ Top 3 drivers – 12, 8 and 21
- ❑ Bottom 3 drivers – 18, 23 and 11



Nearly half of the locations exceed the average cancellation rate

Insights

46% (17) of locations have cancellation rate above average (**23%**). Following locations show a trend of low Driver performance (more cancellations from Drivers) because of low Fair per KM w.r.t. high distance:

- ❑ Anekal – 42.4%
- ❑ C V Raman Nagar – 38.2%
- ❑ Yelahanka – 32.6%
- ❑ Shanti Nagar – 26.7%
- ❑ Vijayanagar – 25.7%
- ❑ Channapatna – 25%

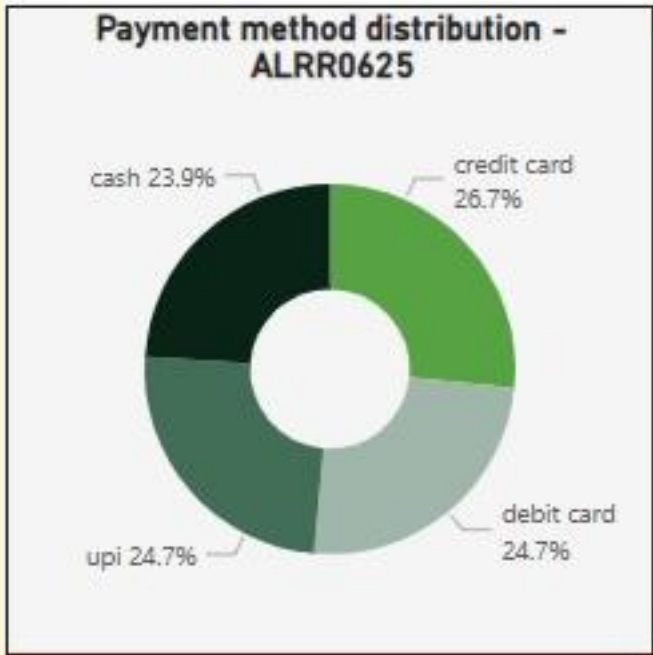
Cancellation rate Vs Other Matrix - ALRR0625						
Assembly	Drivers allotted Trips	Total Cancellations	Trip Cancellation Percent	Trip Rate	Driver Performance	Average of distance
Anekal	33	14	42.4%	34.91	70.4%	17.07
Mahadevapura	43	17	39.5%	59.57	70.3%	13.89
C. V. Raman Nagar	34	13	38.2%	42.57	87.5%	15.71
Doddaballapur	27	10	37.0%	58.98	85.0%	13.59
Yelahanka	43	14	32.6%	50.19	82.9%	15.47
Mahalakshmi Layout	32	10	31.3%	50.10	81.5%	14.12
Pulakeshinagar	30	9	30.0%	60.42	80.8%	12.44
Byatarayanapura	27	8	29.6%	52.24	86.4%	13.24
Hebbal	43	12	27.9%	58.19	86.1%	13.71
Magadi	30	8	26.7%	73.87	78.6%	12.83
Shanti Nagar	30	8	26.7%	50.01	84.6%	15.45
Vijay Nagar	35	9	25.7%	46.10	86.7%	15.94
Channapatna	40	10	25.0%	46.43	88.2%	16.13
Govindraj Nagar	40	10	25.0%	51.01	85.7%	13.61
Kanakapura	36	9	25.0%	49.34	84.4%	14.00
Bangalore South	43	10	23.3%	70.45	86.8%	10.81
Other Assemblies	39	9	23.1%	71.83	90.9%	13.58
Shivajinagar	35	8	22.9%	54.99	93.1%	15.05
Nelamangala	22	5	22.7%	56.50	89.5%	14.76
Devanahalli	31	7	22.6%	52.24	85.7%	15.24
Jayanagar	40	9	22.5%	50.74	96.9%	15.52
Sarvagnanagar	32	7	21.9%	47.59	96.2%	15.45
Bommanahalli	37	8	21.6%	50.25	87.9%	16.00
B. T. M. Layout	33	7	21.2%	50.02	89.7%	12.66
Gandhi Nagar	40	8	20.0%	43.89	86.5%	15.30
Padmanabhanagar	32	6	18.8%	59.31	86.7%	16.14
Basavanagudi	34	6	17.6%	66.29	90.3%	11.32
Chickpet	34	6	17.6%	53.43	90.3%	14.64
Krishnarajapuram	29	5	17.2%	52.94	85.7%	14.31
Dasarahalli	39	6	15.4%	50.87	94.3%	13.74
Ramanagaram	46	7	15.2%	61.87	92.9%	13.69
Hoskote	28	4	14.3%	63.25	88.9%	12.05
Malleshwaram	30	3	10.0%	45.96	93.1%	14.05
Yeshwantpur	40	4	10.0%	56.32	92.3%	12.80
Rajaji Nagar	21	2	9.5%	56.20	100.0%	14.46
Chamrajpet	34	3	8.8%	52.43	96.9%	16.22
Rajarajeshwarinagar	35	3	8.6%	44.09	94.1%	15.04
Total	1277	294	23.0%	53.11	87.8%	14.39



Encourage customers to opt for Digital payment methods

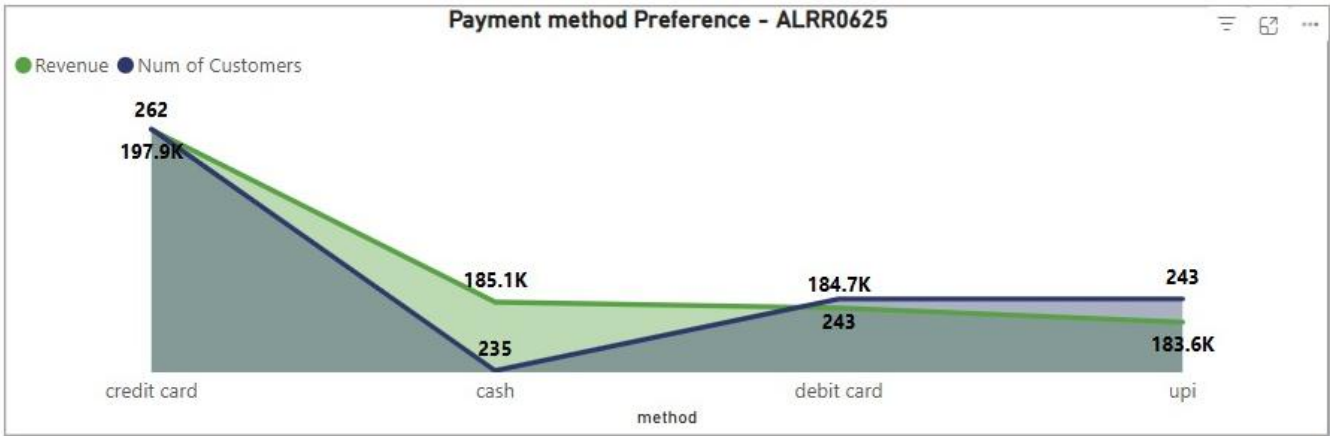
Insights

Digital transactions – **76%** (Credit Card > Debit Card > UPI) outperformed Cash transactions – **24%**.



Insights

Unusual revenue distribution — **185.1K of revenue** for **235 rides** suggests that while fewer customers use cash, their trips tend to be high-value rides



Recommendations

❑ Revenue generation

- ❑ Offer targeted discounts or cashbacks for rides during Early Morning, Evening, Night, and Midnight windows.
- ❑ Time slots between 4 PM and 8 PM show higher average fares but generate relatively lower revenue indicating underutilization from both operational and marketing standpoints

❑ Drivers' performance

- ❑ Driver incentivization program to improve performance

❑ Rides cancellation

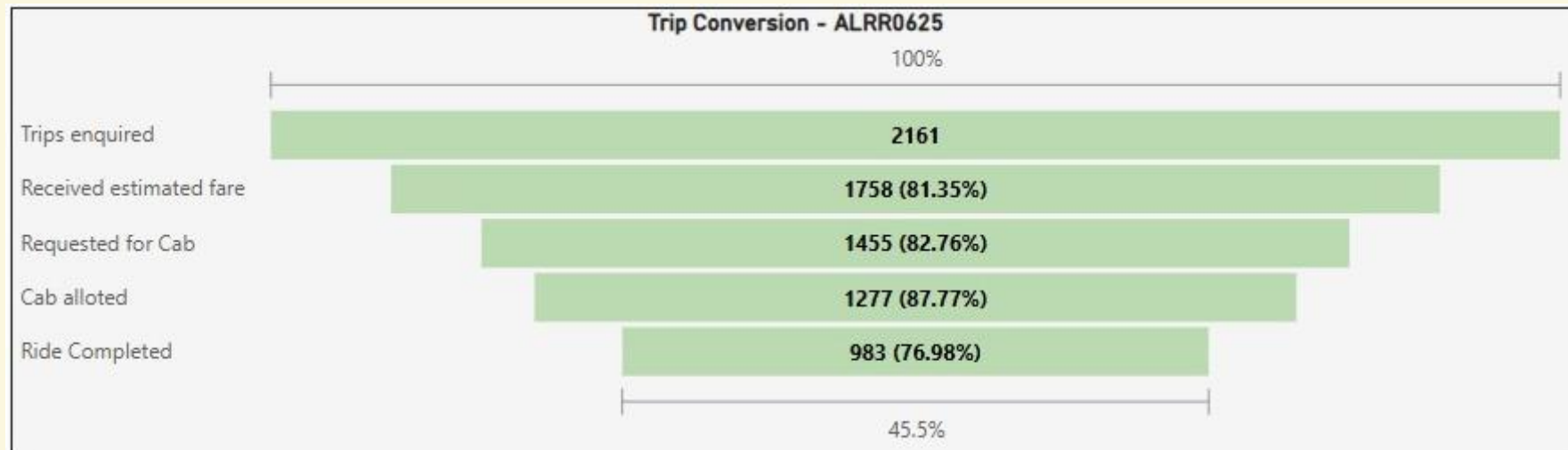
- ❑ Addressing driver and customer cancellations can improve ride completion to 59%.

❑ Payment methods

- ❑ Encourage Digital Alternatives for High-Value Rides- Offering exclusive discounts or smoother checkout for high-value transactions on Digital payment methods

❑ Ride conversion rate

- ❑ Reducing wait-times by better driver allocation can further improve the completion rate to 67%
- ❑ Experiment with A/B tests for ride suggestion algorithms and booking flows to increase conversions to 80%



Appendix – Data Source

Tables information

The dataset comprises five tables: **Assembly** - storing assembly zone details, **Duration** - providing time period identifiers, **Payment** - detailing payment methods, **Trip Details** - capturing trip request and status information and **Trips** - containing core trip data like fare, locations, and IDs.

Variables information

This data dictionary outlines key variables for analyzing trip data. It includes unique identifiers for assemblies, durations, payments, and trips. Details cover trip requests, price estimates, driver searches, cancellations, OTP entry, ride completion, fare information, locations, and involved parties like drivers and customers.



Appendix – Data Methodology & Data Assumptions

Data Methodology

We conducted a thorough analysis of the trip data and trip details fact data. The process included:

- ❑ Analyzed data by applying logical filters and used Matrix table to establish various logics for creating measures
- ❑ Using Column quality and Column distribution options, analytically tried to explore errors, duplicates values and data type of each column in all the tables. Ensured relevant headers are promoted in each table before loading to Model view.
- ❑ We identified two tables' Trips and Trips_Details as fact tables and Assembly, Duration and Payment tables as dimension tables. We created fact constellation schema, where we both the facts tables are sharing the data from one or many dimension tables
- ❑ We established one-to-many cardinality between the tables to ensure data integrity, allowing for accurate data aggregation and meaningful analysis

Data Assumptions

- ❑ Conversation rate metric calculates the % of rides that successfully completed out of total number of inquiries (searches) i.e. *Sum of end_rides / Sum of searches*
- ❑ A trip is considered as cancelled if ride is not completed after driver allotment i.e. *sum of Searches_got_quote minus sum of end_rides*
- ❑ For each TripId, distance covered by the driver is considered as optimized distance

